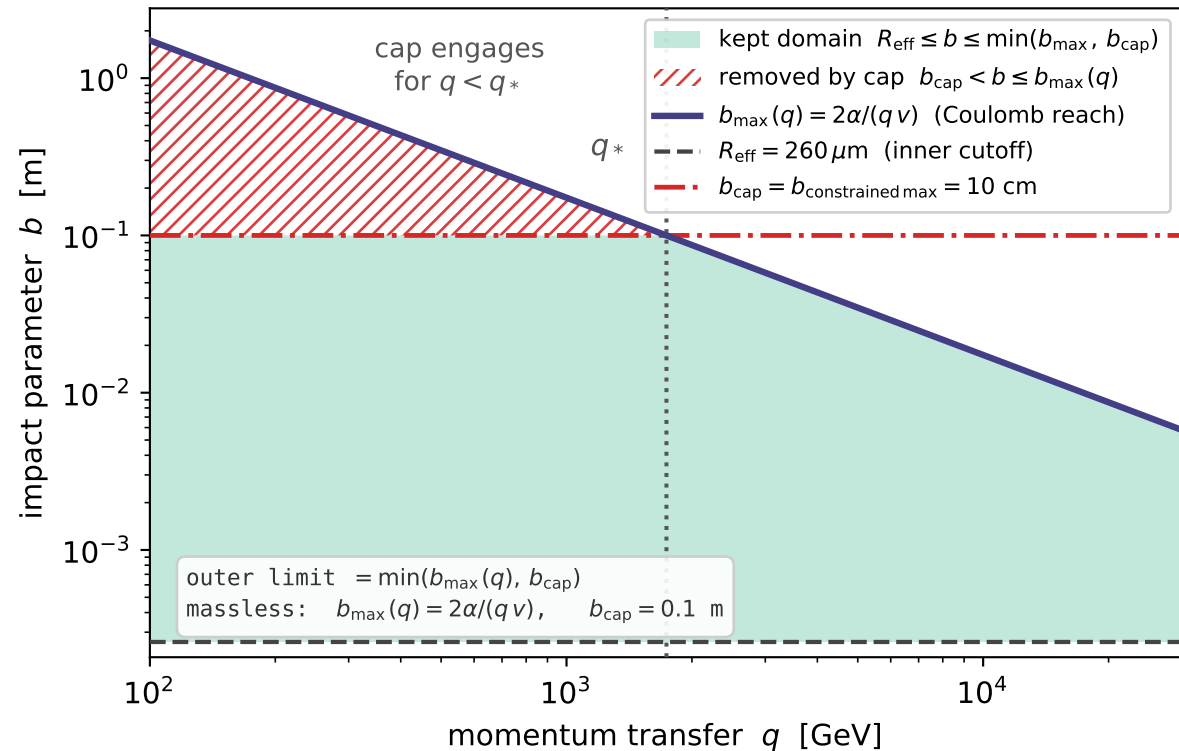
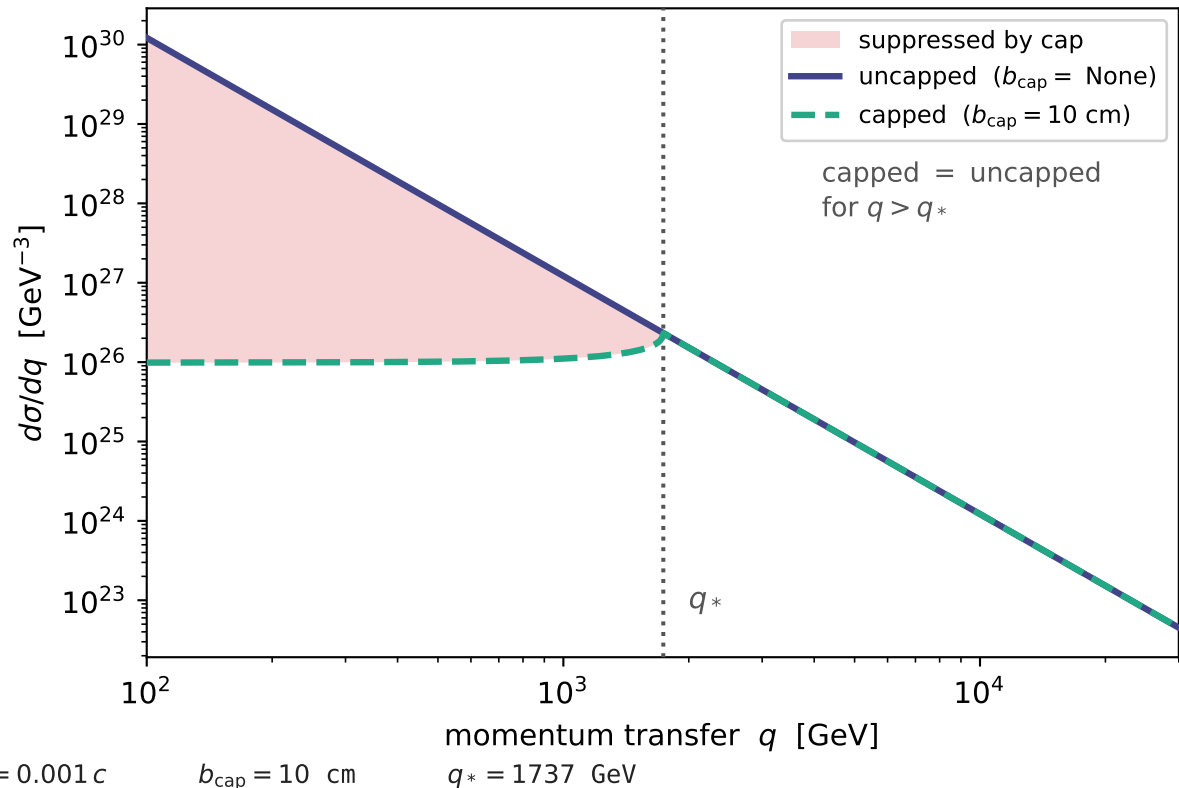


Impact-parameter cap: discarding the large- b wedge ($b > b_{\text{cap}}$) for $q < q_*$ suppresses $d\sigma/dq$

A. Integration domain in the (q, b) plane



B. Effect on $d\sigma/dq$



$$\alpha = \alpha_n N_n = 4.40 \times 10^{14} \quad (\alpha_n = 4.1 \times 10^{-6}) \quad v = 0.001 c$$

$$b_{\text{cap}} = 10 \text{ cm}$$

$$q_* = 1737 \text{ GeV}$$

Massless / Coulomb slice at the real 10 cm production cap $b_{\text{constrained max}} = 0.1 \text{ m}$. Finite- λ slices are affected only where the reach exceeds 10 cm (strong coupling / long range).